

Seat No.: _____

Stony Brook Institute at Anhui University

2022 Fall Semester

Fundamentals of Programming Final Exam (Paper B)

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Multiple Choices (10 items, 1 point each)

Score

(Choose the best answer for each. The answers **must** be written in the form below.)

No.	1	2	3	4	5	6	7	8	9	10
Answer										

- The origin (0, 0) is at the () in the coordinate system for pixels in a computer window.
A. top left B. bottom left C. top right D. center
- Which of the following characters cannot appear in a variable name?
A. letters B. numbers C. underscore D. brackets
- Which of the following is a character constant?
A. "m" B. "!" C. a D. \$
- The value of the expression $5*6\%4+13/3*2$ is ().
A. 8 B. 10 C. 12 D. 14
- Which of the following assignment statements is right?
A. double d=(int)2; B. int i=3.14; C. int i=(double)4; D. char c="a";
- Which of the following is a legal logical expression?
A. (7+8)&&(9-5) B. (9*5)||(9*7) C. (9>6)&&(8<10) D. (9%4)||(8*3)
- Which of the following statements about the for loop is right?
A. The for loop body can contain multiple statements, but must be enclosed in curly braces.
B. The for loop can only be used for a fixed number of iterations.
C. In the for loop, you cannot use the break statement to jump out of the loop.
D. The for loop executes the loop body statements first, and then judges the loop conditional test.

8. Which of the following keywords is used to create an object?
A. create B. class C. setup D. new
9. How to access the last element in an array named a?
A. a[a.length()-1] B. a[a.length()] C. a[a.length-1] D. a[a.length]
10. Which of the following is to judge whether two strings are equal?
A. s1 == s2 B. s1 != s2 C. s1.equals(s2) D. s1.compareTo(s2)

II. Blank Fillings (10 items, 2 points each)

Score	
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(The answers **must** be written in the blanks below.)

- (1) _____ (2) _____
- (3) _____ (4) _____
- (5) _____ (6) _____
- (7) _____ (8) _____
- (9) _____ (10) _____

1. // Add to each number the number that follows in the array, skip the last value.

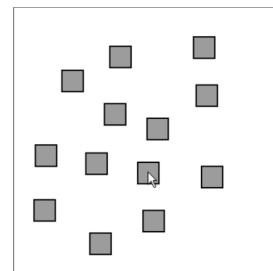
```
int[] nums = {5, 4, 2, 7, 6, 8, 5, 2, 8, 14};
for (int i = 0; _____ (1) _____; i++) {
    nums[i] += _____ (2) _____;
}
printArray(nums);
```

2. // Adding squares with side length of 20 pixels whenever the mouse is pressed.

```
void setup() {
    size(200, 200);
    background(255);
}
```

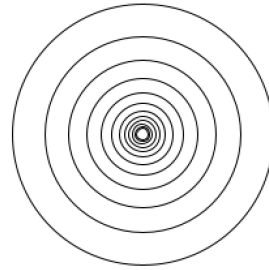
```
void draw() { }
```

```
void _____ (3) _____ () {
    stroke(0);
    fill(175);
    rectMode(CENTER);
    _____ (4) _____;
}
```



3. // Write a recursive function to draw a series of concentric circles.

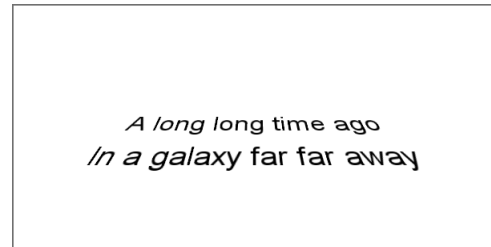
```
void drawCircle(int x, int y, float diam) {
    _____(5)_____;
    if (diam > 10)
    {
        diam *= 0.75;
        _____(6)_____;
    }
}
```



4. // Display text that is centered and rotated 45 degrees to appear floating.

```
PFont f;
void setup() {
    size(400, 200, _____(7)_____);
    f = createFont("Arial", 16);
}
```

```
void draw() {
    background(255);
    fill(0);
    translate(width/2, height/2);
    _____(8)_____;
    textFont(f);
    textAlign(CENTER);
    text("A long long time ago\nIn a galaxy far far away", 0, 0);
}
```



5. // Displaying the pixels of an image with width and height of 200 pixels.

```
size(200, 200);
PImage img = _____(9)_____("flower.jpg");
loadPixels();
for (int y = 0; y < height; y++) {
    for (int x = 0; x < width; x++) {
        int loc = _____(10)_____;
        pixels[loc] = img.pixels[loc];
    }
}
updatePixels();
```

III. Corrections (5 items, 2 points each)

Score	
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1.

```
size(200, 200);
background(255);
for (y=0; y<height; y+=10) {
    line(0, y, width, y);
}
```
2.

```
String message = "Happy New Year!";
int len = message.length;
print("This message has " + len + " letters.");
```
3.

```
class Car {
    int x, y;
    color c;
    float speed;

    void Car(int _x, int _y, int _c) {
        x = _x;
        y = _y;
        c = _c;
        speed = 0.0;
    }
}
```
4.

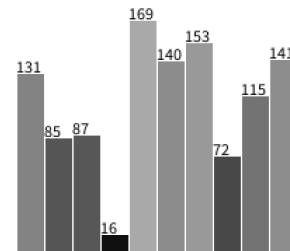
```
Car[] cars = new Car[18];
for (int i = 0; i < Car.length; i++)
{
    cars[i] = new Car(0, i*10, color(random(255)));
}
```
5.

```
size(200, 200);
loadPixels();
for (int i = 0; i <= pixels.length; i++) {
    float rand = random(255);
    pixels[i] = color(rand);
}
updatePixels();
```

IV. Programming (5 items, 60 points in total)

Score	
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1. There are some comma-separated numbers from a text file named "data.txt", such as "131,85,87,16,169,140,153,72,115,141". Program to generate a simple data visualization showed as the right-down figure. (10 points)



2. When pixels differ greatly from their neighbors, they are most likely "edge" pixels. Program to implement a simple vertical edge detection algorithm for a 200×200 pixels image file named "sunflower.jpg". (10 points)

3. Program to display live video through a computer's camera by using the Processing video library. (10 points)

4. Program to implement scrolling headlines, which should display a different headline in 16-point size Arial font each time the previous headline leaves the screen. The headlines are as follows: "I like Processing.", "I like Java, too.", "Do you like Processing and Java?". (15 points)

5. Using object-oriented programming to simulate 100 bouncing balls, which can bounce around in the horizontal direction, and the position, size, color and speed of each ball are different. (15 points)